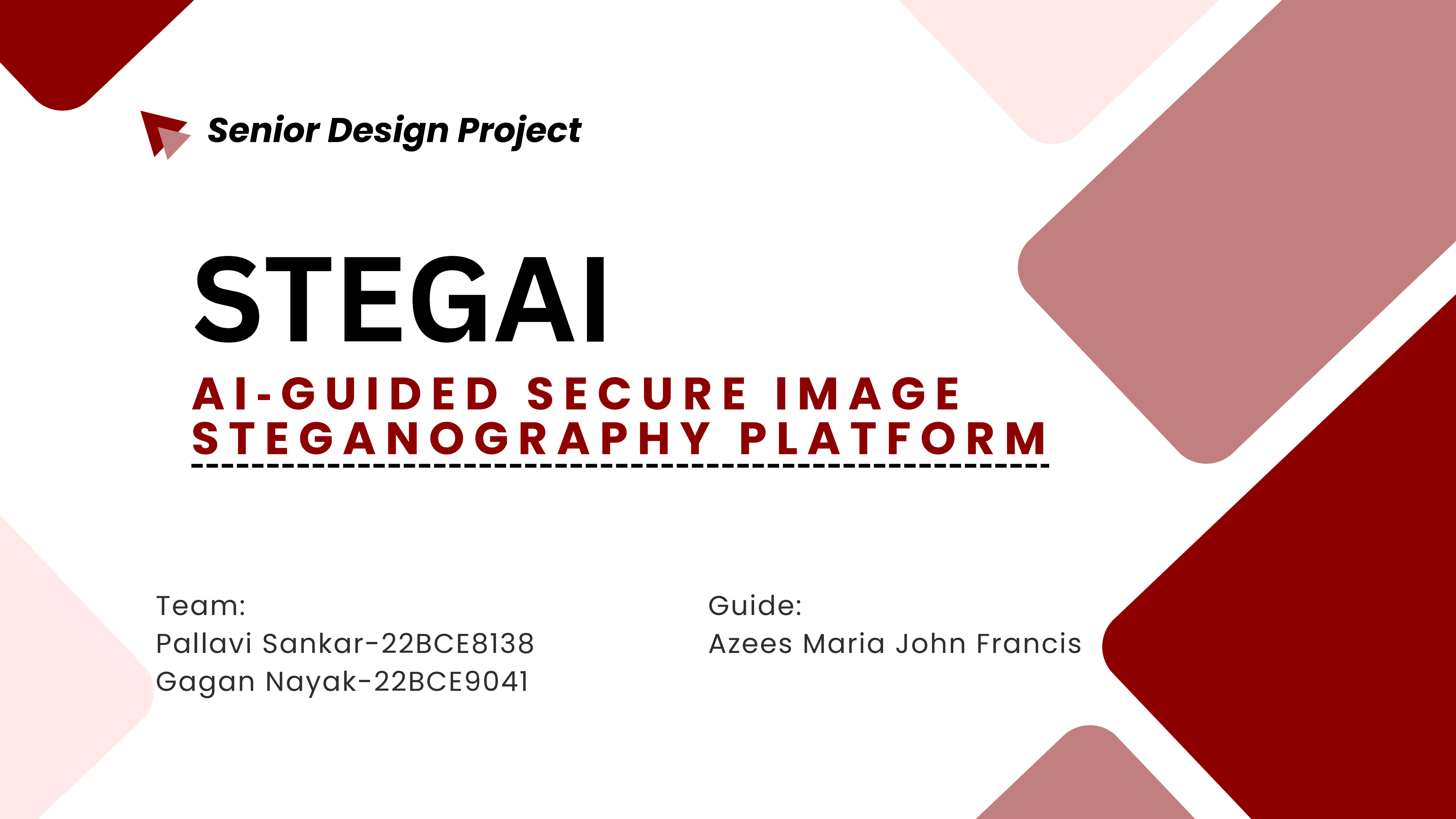




 **Senior Design Project**

STEGAI

**AI-GUIDED SECURE IMAGE
STEGANOGRAPHY PLATFORM**

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INTRODUCTION

Digital communication systems require both security and privacy to ensure safe information exchange. While cryptography effectively maintains data confidentiality, it does not conceal the existence of the communication itself, making encrypted transmissions visible and potentially suspicious. To address this, steganography enables covert communication by embedding data within images. However, traditional steganographic techniques often introduce detectable patterns, highlighting the need for more intelligent and adaptive approaches.



ELLIPTIC CURVE CRYPTOGRAPHY(ECC)

The graph represents an elliptic curve defined by an equation of the form $y^2 = x^3 + ax + b$.

Points such as P, Q, and R lie on this curve. ECC works by performing operations like point addition and scalar multiplication on these points. For example, adding two points on the curve results in another point on the same curve. These mathematical operations are easy to compute in one direction but extremely difficult to reverse, which forms the basis of ECC's security.

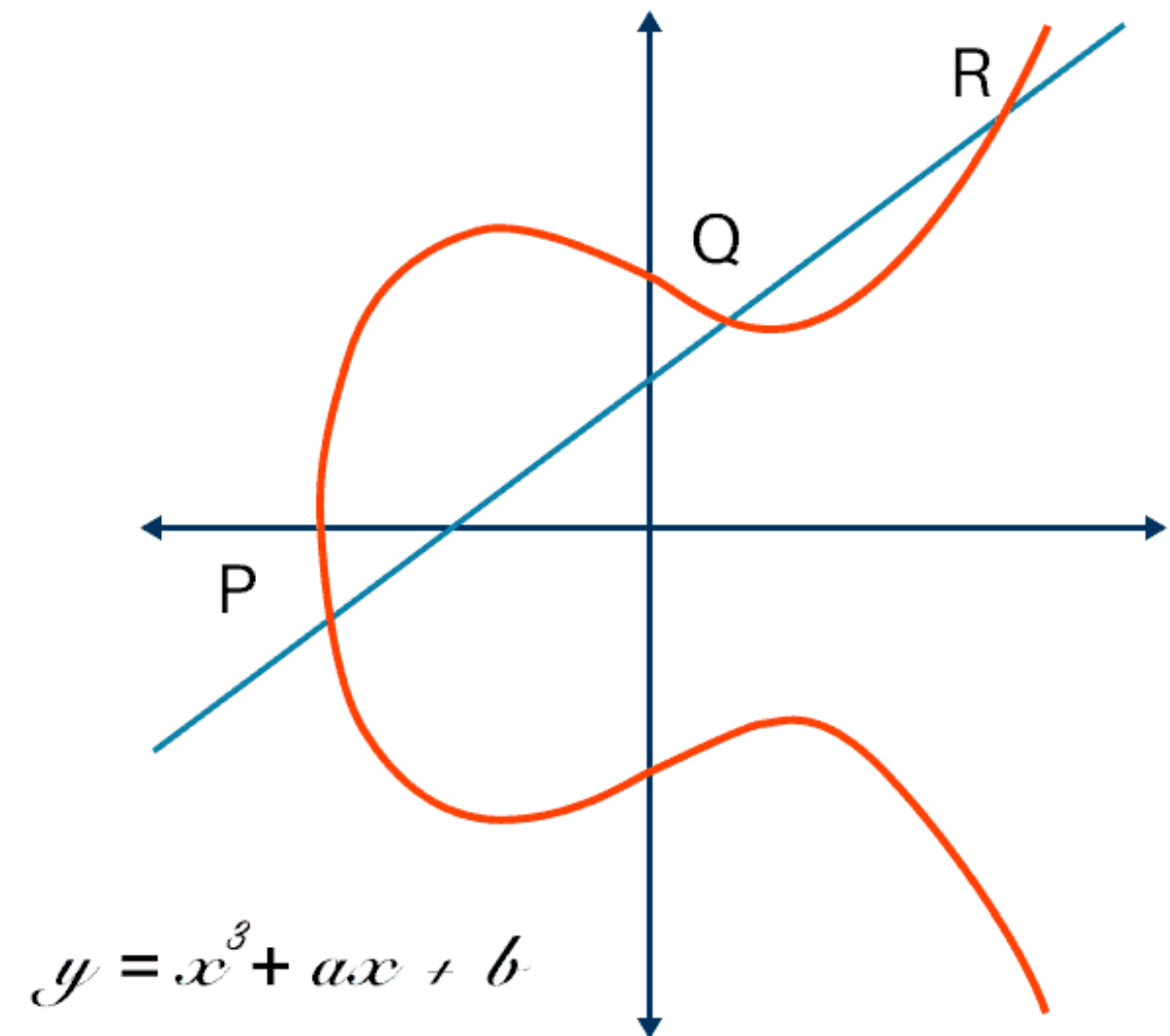
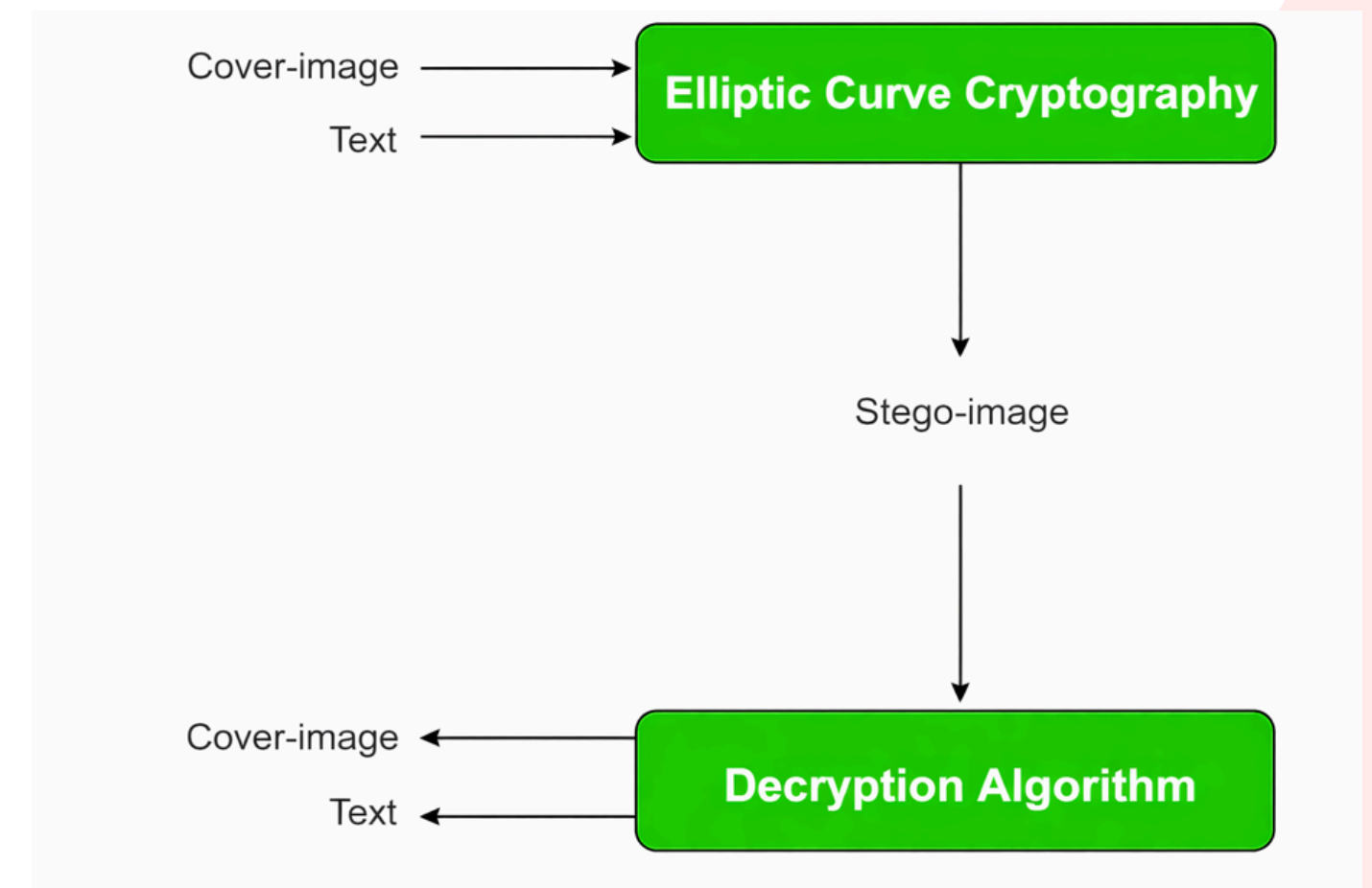


IMAGE STEGANOGRAPHY

In this system, the cover image and the secret text are first given as inputs to the Elliptic Curve Cryptography (ECC) module. ECC encrypts the secret text using a public key, converting it into an unreadable encrypted form. The encrypted data is then embedded into the cover image, resulting in a stego image that looks similar to the original image.

At the receiver side, the stego image is passed through the ECC decryption process using the corresponding private key. The encrypted data is extracted and decrypted to recover the original secret text, while the cover image remains visually unchanged.



Problem Statement & **OBJECTIVES**

- Traditional LSB embedding is **uniform across pixels**
- This creates **predictable statistical patterns**
- Detectable using:
 - Histogram analysis
 - Chi-square tests
 - SPAM / SRM models
- No consideration of:
 - Image texture
 - Region complexity
 - Safe payload limits
- Hidden communication becomes **detectable even if encrypted**

- Encrypt message using ECC (content security)
- Use AI to decide where to embed data
- Control payload capacity to avoid over-embedding
- Minimize statistical distortion in images
- Build a complete web-based system

LITERATURE SURVEY

Authors & Citation	Contribution	Proposed Work	Advantages	Disadvantages
Pevný, T., Bas, P., Fridrich, J., 2010, Steganalysis by Subtractive Pixel Adjacency Matrix (SPAM), IEEE TIFS Published: 2010	Introduced SPAM features to detect steganographic artifacts using pixel adjacency statistics	Feature-based steganalysis using pixel difference and adjacency matrices	Effective against LSB and spatial-domain steganography; relatively low computational complexity	Limited effectiveness against modern adaptive steganography; vulnerable to advanced distortion-minimization techniques
Fridrich, J., Kodovský, J., 2012, Rich Models for Steganalysis of Digital Images, IEEE TIFS Published: 2012	Proposed rich statistical feature models for high-accuracy steganalysis	Spatial Rich Models (SRM) using high-dimensional residual-based features	Very high detection accuracy; strong benchmark for steganalysis research	High computational cost; large feature dimensionality; not suitable for real-time systems
Yang, Z., Zhang, Y., Li, Y., 2023, IStego100K: A Large-Scale Image Steganalysis Dataset, IEEE Access Published: 2023	Introduced a large-scale benchmark dataset for training and evaluating image steganalysis models	Introduced a large-scale benchmark dataset for training and evaluating image steganalysis models	Large dataset size; diverse embedding methods; supports evaluation of low- and mid-level steganalysis techniques	Focuses on detection rather than embedding; high storage and computational requirements; not directly applicable for lightweight systems

Holub, V., Fridrich, J., Denemark, T., 2014, Universal Distortion Function for Steganography in an Arbitrary Domain, IEEE TIFS Published: 2014	Introduced a universal distortion framework for content-adaptive steganography	Distortion-minimizing adaptive embedding across multiple domains	Reduced detectability under statistical steganalysis; adaptable to various embedding domains	Computationally intensive; complex distortion modeling; not designed for low-complexity systems
Apau, R., Badu, K., Amankwah, R., 2025, Image Steganography Techniques for Resisting Statistical Steganalysis Attacks, Scientific Reports Published: 2025	Provided a comprehensive survey of techniques aimed at resisting statistical steganalysis attacks	Analysis of adaptive, content-aware, and distortion-based embedding methods	Clearly categorizes statistical attacks; supports adaptive embedding strategies; directly relevant to covert communication	Survey-based; lacks implementation details; no unified embedding algorithm proposed
De La Croix, N. J., 2024, Comprehensive Survey on Image Steganalysis Using Deep Learning, Results in Engineering Published: 2024	Reviewed deep learning-based steganalysis techniques and their effectiveness	Survey of CNN-based and hybrid steganalysis models	Explains evolution from statistical to deep learning steganalysis; useful for defining high-level attacks	Focuses on detection rather than defense; requires large datasets; high computational complexity

EXISTING SYSTEM VS PROPOSED SYSTEM

- Uses basic LSB embedding
- Data embedded uniformly across pixels
- Ignores image characteristics (texture, complexity)
- Causes statistical distortion
- Easily detected by steganalysis

“Proposed system introduces content-aware adaptive embedding instead of uniform embedding.”

- Uses ECC encryption + adaptive steganography
- The input image is divided into **8×8 blocks for analysis**
- Each block is evaluated using:
 - Entropy (randomness)
 - Variance (texture)
- Based on this analysis:
 - High-texture regions are selected as **safe embedding zones**
 - Smooth regions are avoided
- Encrypted data is embedded using **adaptive LSB**, where:
 - Embedding locations are controlled
 - Payload is adjusted based on region suitability
- This approach reduces:
 - Statistical distortion
 - Detectability under steganalysis

THREAT MODEL AND ATTACK ANALYSIS

The proposed system primarily addresses low-level and mid-level steganalysis attacks by minimizing statistical distortions through AI-guided adaptive embedding. By intelligently selecting embedding regions and controlling payload capacity, the system reduces detectability under common statistical and feature-based attacks.

Low-Level Attacks:

LSB Statistical Analysis

Looks for unnatural randomness in the least significant bits of pixels.

Histogram Analysis

Compares pixel intensity distributions to find anomalies caused by embedding.

Chi-Square Attack

Tests whether pixel pairs follow natural probability distributions.

Noise Variance Analysis

Detects increased noise levels caused by data hiding.

Mid-Level Attacks:

SPAM-Based Steganalysis

Analyzes pixel adjacency and difference patterns.

Texture Inconsistency Detection

Identifies abnormal texture transitions across image regions.

Block Artifact Analysis

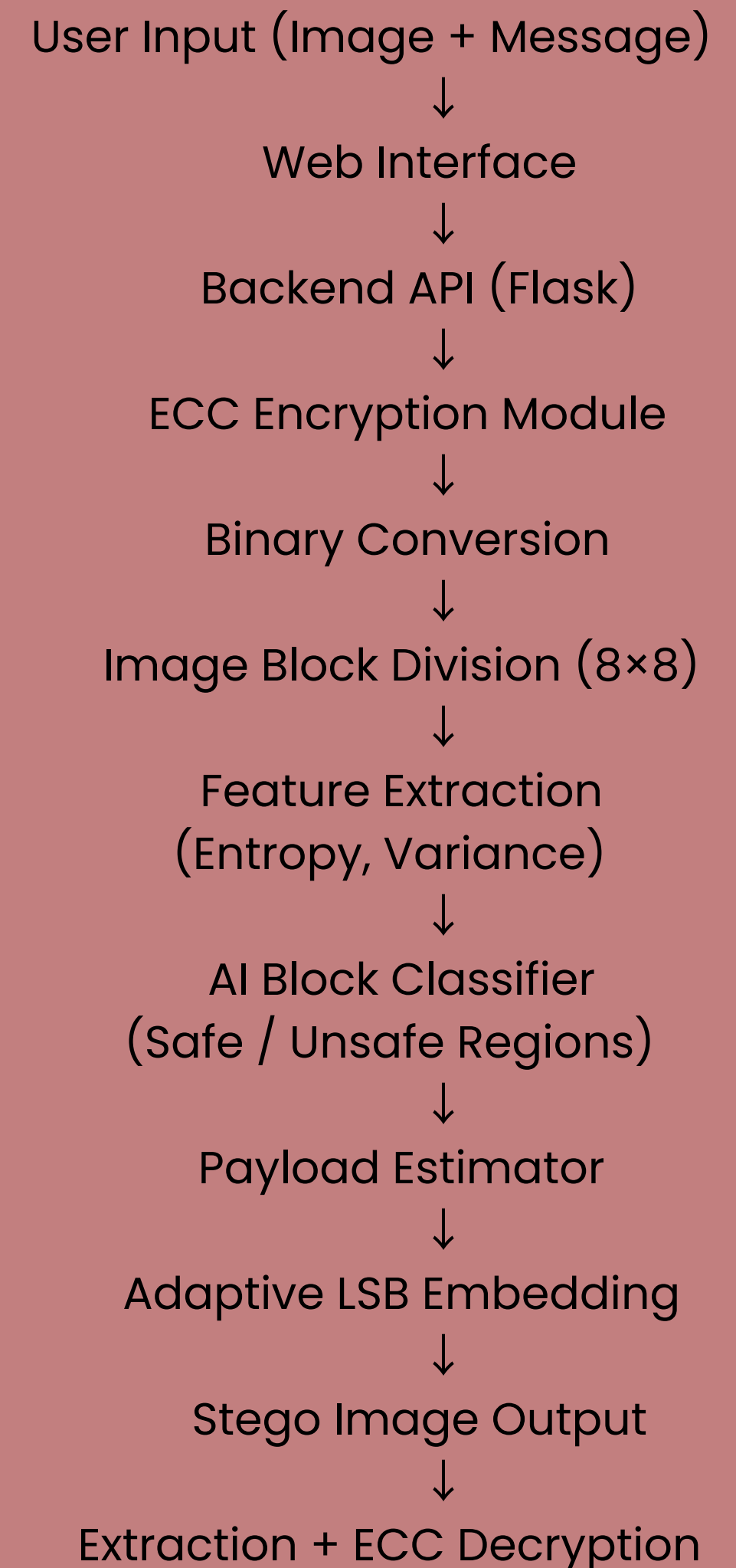
Detects unnatural differences between neighboring image blocks.

Feature-Based Machine Learning Attacks

Uses entropy, variance, and co-occurrence features with classifiers like SVM or Random Forest.

System Architecture

- User inputs image and message via web interface
- Backend encrypts message using ECC
- Image divided into blocks for feature analysis
- AI selects safe regions for embedding
- Adaptive LSB embeds encrypted data
- Stego image generated and later extracted + decrypted



TECHNOLOGY STACK



- Frontend: HTML, CSS, JavaScript



- Backend: Flask-Python



- Image Processing: OpenCV, Numpy



- Machine Learning: scikit-learn



ENCRYPTION

- Encryption: AES+ECC



EVALUATION

- Evaluation: PSNR, SSIM

MODULES

- Image Analysis Module
 - Divides image into blocks and computes entropy and variance
 - Identifies suitable embedding regions
- ECC Encryption Module
 - Encrypts message before embedding, ensuring data security even if extracted
- Adaptive Embedding Module
 - Embeds data only in safe regions
 - Controls bits per pixel dynamically
- Extraction & Decryption Module
 - Extracts embedded bits
 - Decrypts using ECC to recover message
- Evaluation Module
 - Measures PSNR and SSIM
 - Evaluates payload vs distortion

AI MODULE

- AI module analyzes image at block level (8×8 regions)
- Extracts key statistical features:
- Entropy → measures randomness
- Variance → measures texture intensity
- Decision process:
- High entropy + high variance → safe for embedding
- Low texture regions → avoided
- Benefits:
- Embedding mimics natural image noise
- Reduces visible and statistical artifacts
- Improves resistance to detection

This reduces statistical artifacts and improves imperceptibility. AI ensures embedding follows image structure instead of random placement. This heuristic approach will be extended using a trained ML classifier (Random Forest)

MLMODULE

- Introduce **Random Forest classifier** for better decision-making
- Input features per block:
 - Entropy
 - Variance
 - (Optional) frequency-domain features
- Output:
 - Classifies blocks as **safe or unsafe**
- Advantages:
 - Eliminates manual threshold tuning
 - Adapts to different image types automatically
 - Improves embedding accuracy and robustness

Replaces rule-based logic with data-driven adaptive embedding.
Model is currently under training and will replace rule-based thresholds.

PROGRESS

- Web-based interface developed with image upload and message input
- ECC encryption and decryption successfully implemented
- Basic LSB embedding and extraction verified
- AI-based feature extraction (entropy and variance) completed
- Initial adaptive embedding logic implemented
- Integration of backend APIs in progress
- Machine learning classifier under development

System is moving from prototype to intelligent adaptive model
Initial end-to-end pipeline tested successfully on sample images.

DEMO

Secure & Covert Communication

Upload Image

Choose File sample_image.jpg



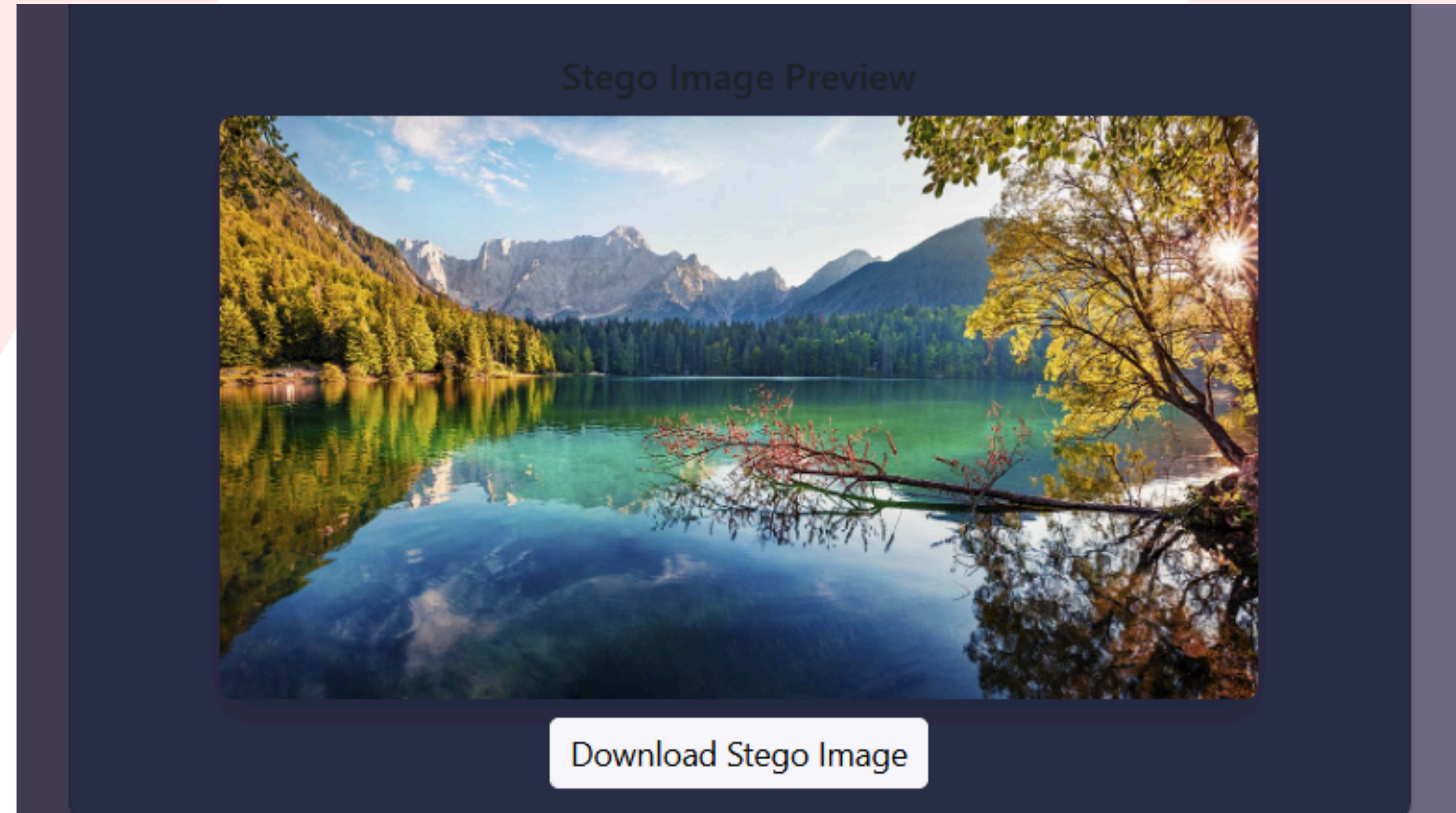
Secret Message

ECC is cool

Encrypt & Hide

Extract Message

Uploading Image and encrypting it with the message.



A preview of the Image shown and the user can download it here.

DEMO

Upload Image

Choose File No file chosen



Secret Message

Encrypt & Hide Extract Message

Extracted Message: ECC is cool

Image with a message uploaded and extracted.

Upload Image

Choose File No file chosen



Secret Message

Encrypt & Hide Extract Message

No hidden message found in this image.

Image with no message is uploaded and extracted.

EVALUATION

- **Image Quality Evaluation**
 - PSNR used to measure noise introduced after embedding
 - SSIM used to measure structural similarity
- **Security Evaluation**
 - Histogram comparison between original and stego images
 - Chi-square analysis for statistical detectability
- **Performance Evaluation**
 - Payload capacity vs image distortion
 - Accuracy of message extraction

Goal is to maintain high image quality while minimizing detectability

Future Plan

- Enhance resistance to deep learning–based steganalysis using adversarial or feedback-guided embedding
- Extend embedding to multiple domains (spatial and frequency) to increase attack complexity
- Implement key-dependent and randomized embedding paths for improved secrecy
- Improve robustness against image processing attacks such as compression, resizing, and noise
- Introduce dynamic payload control to minimize statistical artifacts and avoid over-embedding
- Perform extensive security benchmarking using standard steganalysis datasets and tools

Conclusion

This project presents a secure and intelligent steganography system that combines Elliptic Curve Cryptography with AI-guided adaptive embedding to achieve effective covert communication. By analyzing image characteristics such as entropy and variance, the system selectively embeds data in suitable regions, reducing statistical distortion and improving resistance to common steganalysis techniques. Unlike traditional methods, the proposed approach introduces adaptability and control over embedding, making it more robust and practical. Overall, the system demonstrates a balanced integration of security, intelligence, and usability, making it a strong solution for modern secure communication requirements.

REFERENCE

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- <https://pmc.ncbi.nlm.nih.gov/articles/PMC11404826/>
- <https://www.sciencedirect.com/science/article/pii/S2590005624000195>

The background features two large, light pink geometric shapes. One is a triangle on the left side, and the other is a larger, more complex shape on the right side, both with rounded corners.

THANK YOU